Student Name: _____

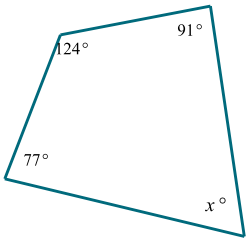
Number of Questions: **24**

Class Name: **ALEKS Integrated Mathematics II**

Instructor Name: **Bodanske, Daniel**

Question 1 of 24

Find the value of x.



Question 2 of 24

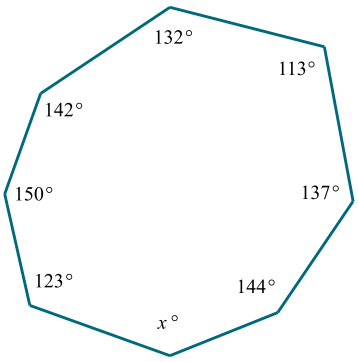
Find the sum of the interior angle measures of a convex decagon (a ten-sided polygon).

Question 3 of 24

The sum of the interior angle measures of a convex polygon is $540°$. How many sides does it have?

Question 4 of 24

The convex polygon below has 8 sides. Find the value of x.



Question 5 of 24

Find the measure of each interior angle and each exterior angle of a regular octagon (an eight-sided polygon).

Question 6 of 24

The measure of each interior angle of a regular polygon is $135°$. Find the number of sides.

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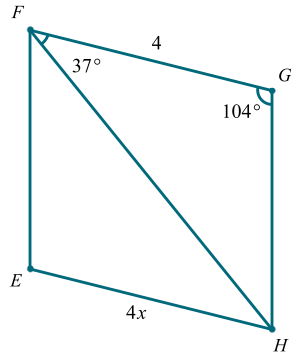
Question 7 of 24

Consider parallelogram $EFGH$ below.

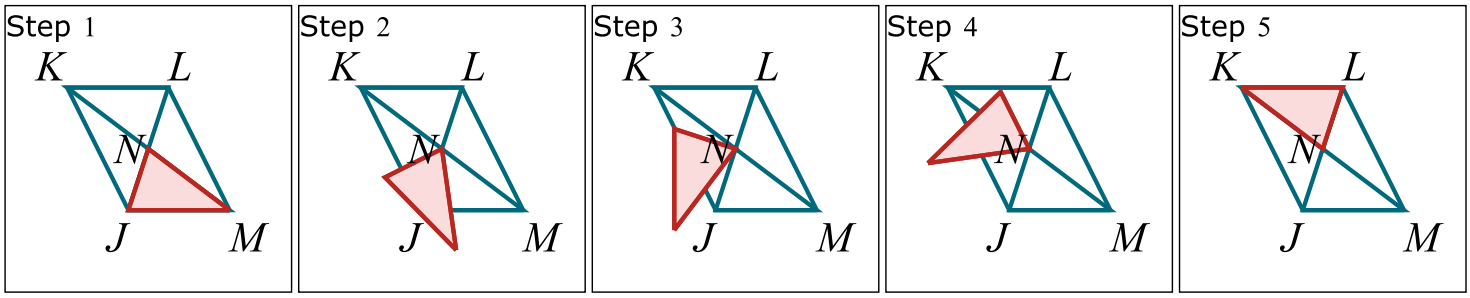
Use the information given in the figure to find $m\angle E$, $m\angle EFH$, and x.



Question 8 of 24

Consider the parallelogram $JKLM$ with diagonals $\overline{JL}$ and $\overline{KM}$, which intersect at point N.

Use the sequence of images to help answer the questions below.



(a) $\triangle JNM$ is congruent to which other triangle?

$\triangle JNM \cong \triangle$ _____

(b) $\overline{JN}$ is congruent to which other segment?

$\overline{JN} \cong$ _____

(c) $\overline{KN}$ is congruent to which other segment?

$\overline{KN} \cong$ _____

(d) The following are all properties of parallelograms. Which one is illustrated by parts (b) and (c)?

- Consecutive interior angles of a parallelogram are supplementary.
- Opposite sides of a parallelogram are congruent.
- Opposite angles of a parallelogram are congruent.
- Alternate interior angles formed by the diagonals of a parallelogram are congruent.
- The diagonals of a parallelogram bisect each other.

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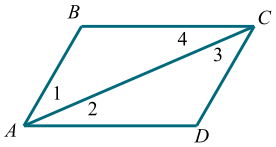
Question 9 of 24

Use the given information to complete the proof of the following theorem.

If a quadrilateral is a parallelogram, then its opposite sides are congruent.

By definition, a parallelogram is a quadrilateral in which both pairs of opposite sides are parallel. Use this definition in your proof.



Given: $ABCD$ is a parallelogram

Prove: $\overline{AB} \cong \overline{CD}$

$\overline{AD} \cong \overline{BC}$

	Statement	Reason
1	$ABCD$ is a parallelogram	Given
2	$\overline{AB} \parallel$	Definition of a Parallelogram
3	$\overline{AD} \parallel$	Definition of a Parallelogram
4	$\angle 1 \cong \angle$	_____
5	$\angle 2 \cong \angle$	_____
6	$\cong$	Reflexive Property
7	$\triangle ABC \cong \triangle$	_____
8	$\overline{AB} \cong \overline{CD}$ $\overline{AD} \cong \overline{BC}$	_____

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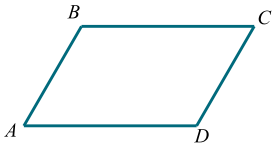
Question 10 of 24

Use the given information to complete the proof of the following theorem.

If opposite angles of a quadrilateral are congruent, then it is a parallelogram.

By definition, a parallelogram is a quadrilateral in which both pairs of opposite sides are parallel. Use this definition in your proof.



Given: $m\angle A = m\angle C$

$m\angle B = m\angle D$

Prove: $ABCD$ is a parallelogram

	Statement	Reason
1	$m\angle A = m\angle C$	Given
2	$m\angle B = m\angle D$	Given
3	$m\angle A + m\angle B + m\angle C + m\angle D = 360°$	Quadrilateral Angle-Sum Property
4	$m\angle A + m\angle B + m\angle \square + m\angle \square = \square°$	Substitution Property
5	$2(m\angle \square + m\angle \square) = \square°$	Simplifying
6	$m\angle A + m\angle B = 180°$	Multiplication and Division Properties
7	$\angle \square$ and $\angle \square$ are supp. angles	Definition of Supplementary Angles
8	$\overline{AD} \parallel \overline{\square}$	_____
9	$m\angle A + m\angle D = \square°$	Substitution Property
10	$\angle A$ and $\angle D$ are supp. angles	Definition of Supplementary Angles
11	$\overline{AB} \parallel \overline{\square}$	_____
12	$ABCD$ is a parallelogram	_____

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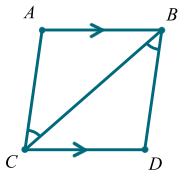
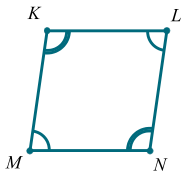
Question 11 of 24

Given the information marked on the figures below, classify each quadrilateral as a "Parallelogram" or "Not necessarily a parallelogram."

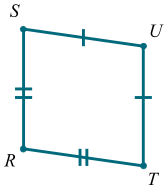
Note that each figure is drawn like a parallelogram, but you should not rely on how the figure is drawn in determining your answers.



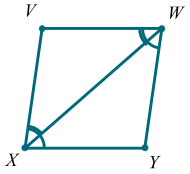
___ Parallelogram
___ Not necessarily a parallelogram



___ Parallelogram
___ Not necessarily a parallelogram



___ Parallelogram
___ Not necessarily a parallelogram



___ Parallelogram
___ Not necessarily a parallelogram

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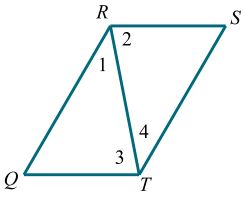
Question 12 of 24

Use the given information to complete the proof of the following theorem.

If opposite sides of a quadrilateral are congruent, then it is a parallelogram.

By definition, a parallelogram is a quadrilateral in which both pairs of opposite sides are parallel. Use this definition in your proof.



Given: $\overline{QR} \cong \overline{ST}$

$\overline{QT} \cong \overline{RS}$

Prove: $QRST$ is a parallelogram

	Statement	Reason
1	$\overline{QR} \cong \overline{ST}$	Given
2	$\overline{QT} \cong \overline{RS}$	Given
3	$\overline{\quad} \cong \overline{\quad}$	Reflexive Property
4	$\triangle \quad \cong \triangle \quad$	_____
5	$\angle 1 \cong \angle \quad$	CPCTC Property
6	$\angle 2 \cong \angle \quad$	CPCTC Property
7	$\overline{QR} \parallel \overline{\quad}$	_____
8	$\overline{QT} \parallel \overline{\quad}$	_____
9	$QRST$ is a parallelogram	_____

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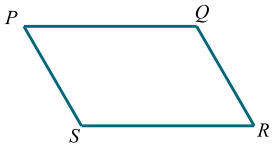
Question 13 of 24

Use the given information to complete the proof of the following theorem.

If a quadrilateral is a parallelogram, then its opposite angles are congruent.

By definition, a parallelogram is a quadrilateral in which both pairs of opposite sides are parallel. Use this definition in your proof.



Given: $PQRS$ is a parallelogram

Prove: $\angle Q \cong \angle S$

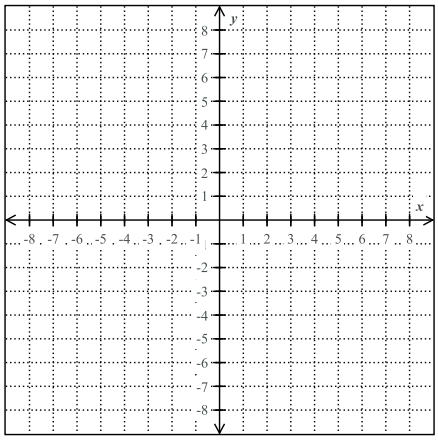
$\angle P \cong \angle R$

	Statement	Reason
1	$PQRS$ is a parallelogram	Given
2	$\overline{PQ} \parallel$ ____	Definition of a Parallelogram
3	$\overline{PS} \parallel$ ____	Definition of a Parallelogram
4	$\angle P$ and $\angle$ ____ are supp. angles	If lines $\parallel$, then cons. int. $\angle$s are supp.
5	$\angle P$ and $\angle$ ____ are supp. angles	If lines $\parallel$, then cons. int. $\angle$s are supp.
6	$\angle Q \cong \angle$ ____	_____
7	$\angle R$ and $\angle$ ____ are supp. angles	_____
8	$\angle P \cong \angle$ ____	_____

Question 14 of 24

Draw the polygon whose vertices are $(-1, -2)$, $(4, -3)$, $(4, 3)$, and $(-1, 0)$.

Then choose the best description of the polygon from the choices below.



Square	Triangle	Trapezoid	None of these
○	○	○	○

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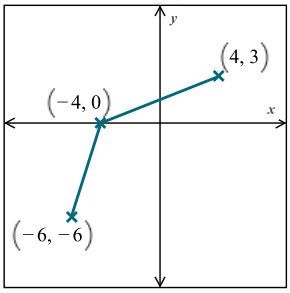
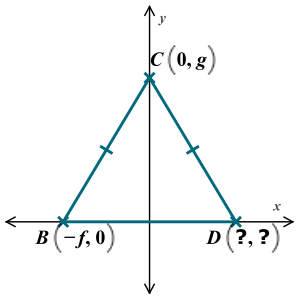
Question 15 of 24

Three vertices of a parallelogram are shown in the figure below.
Give the coordinates of the fourth vertex.



Question 16 of 24

Isosceles triangle BCD is shown below. Give the coordinates of D.



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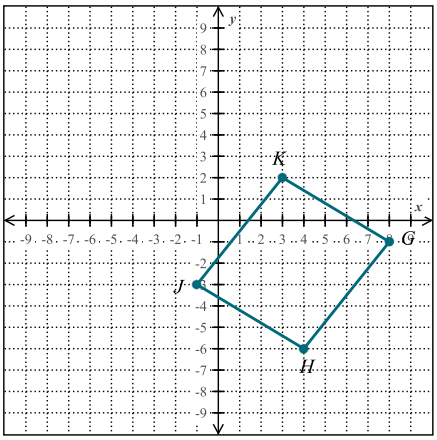
Question 17 of 24

Consider quadrilateral $GHJK$ below.



Note that $GHJK$ has vertices $G(8,\ -1)$, $H(4,\ -6)$, $J(-1,\ -3)$, and $K(3,\ 2)$.
Complete the following to determine if $GHJK$ is a parallelogram.

(a) Find the slope of $\overline{GK}$ and the slope of $\overline{HJ}$.

Slope of $\overline{GK}$:

Slope of $\overline{HJ}$:

(b) Find the slope of $\overline{JK}$ and the slope of $\overline{GH}$.

Slope of $\overline{JK}$:

Slope of $\overline{GH}$:

(c) From parts (a) and (b), what can we conclude?

- The quadrilateral is not a parallelogram because it has a pair of opposite sides that are not parallel.
- The quadrilateral is a parallelogram because it has one pair of opposite sides that are parallel, even though the other two sides are not parallel.
- The quadrilateral is a parallelogram because it has two pairs of opposite sides that are parallel.
- It cannot be determined if the quadrilateral is a parallelogram.

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Question 18 of 24

Answer the questions about the figures below.

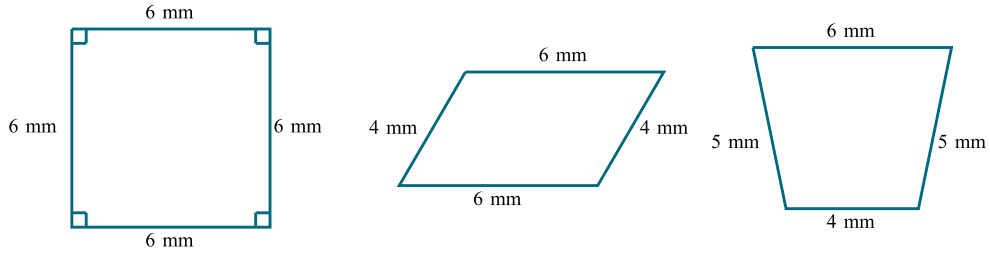


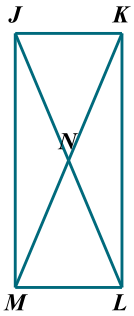
Figure A Figure B Figure C

(a) Which figures are parallelograms? Mark all that apply. ☐ Figure A ☐ Figure B ☐ Figure C ☐ None of the figures
(b) Which figures are squares? Mark all that apply. ☐ Figure A ☐ Figure B ☐ Figure C ☐ None of the figures
(c) Which figures are rectangles? Mark all that apply. ☐ Figure A ☐ Figure B ☐ Figure C ☐ None of the figures

Question 19 of 24

In the rectangle below, $JN = 2x + 2$, $KM = 3x + 11$, and $m\angle NJM = 25°$.

Find LN and $m\angle KNL$.



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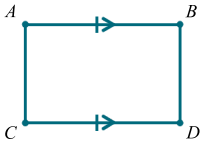
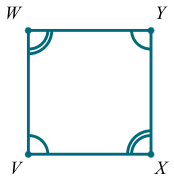
Question 20 of 24

Given the information marked on each figure below, select *all* classifications that must be true.

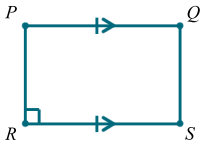
Note that each figure is drawn like a rectangle, but you should not rely on the way the figure is drawn in determining your answers.



___ Quadrilateral
___ Parallelogram
___ Rectangle



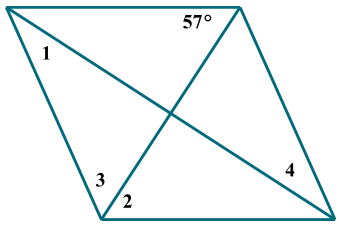
___ Quadrilateral
___ Parallelogram
___ Rectangle



___ Quadrilateral
___ Parallelogram
___ Rectangle

Question 21 of 24

For the rhombus below, find the measures of $\angle 1$, $\angle 2$, $\angle 3$, and $\angle 4$.



$m\angle 1 =$ _____°

$m\angle 2 =$ _____°

$m\angle 3 =$ _____°

$m\angle 4 =$ _____°

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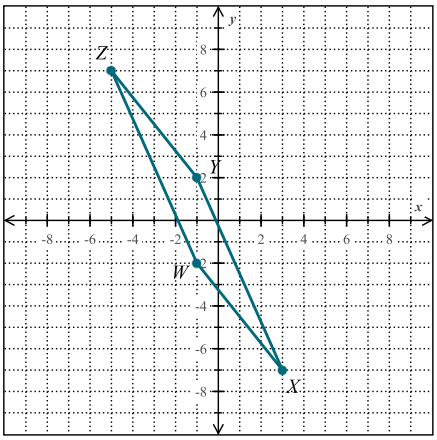
Question 22 of 24

Consider parallelogram $WXYZ$ below.



Note that $WXYZ$ has vertices $W(-1,\ -2)$, $X(3,\ -7)$, $Y(-1,\ 2)$, and $Z(-5,\ 7)$.
Answer the following to determine if the parallelogram is a rectangle, rhombus, square, or none of these.

(a) Find the slope of $\overline{XY}$ and the slope of a side adjacent to $\overline{XY}$.

Slope of $\overline{XY}$: ☐

Slope of side adjacent to $\overline{XY}$: ☐

(b) Find the length of $\overline{XY}$ and the length of a side adjacent to $\overline{XY}$.
Give exact answers (not decimal approximations).

Length of $\overline{XY}$: ☐

Length of side adjacent to $\overline{XY}$: ☐

(c) From parts (a) and (b), what can we conclude about parallelogram $WXYZ$? Check all that apply.

- ☐ $WXYZ$ is a rectangle.
- ☐ $WXYZ$ is a rhombus.
- ☐ $WXYZ$ is a square.
- ☐ $WXYZ$ is none of these.

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Question 23 of 24

For each of the following quadrilaterals, select *all* the properties that must be true.

(a) Which properties must be true for parallelograms?

- [ ] Four right angles
- [ ] All sides congruent
- [ ] Two pairs of parallel sides
- [ ] Only one pair of parallel sides

(b) Which properties must be true for rectangles?

- [ ] Four right angles
- [ ] All sides congruent
- [ ] Two pairs of parallel sides
- [ ] Only one pair of parallel sides

(c) Which properties must be true for rhombuses?

- [ ] Four right angles
- [ ] All sides congruent
- [ ] Two pairs of parallel sides
- [ ] Only one pair of parallel sides

Question 24 of 24

True or False?

a. Every rhombus is a parallelogram.

b. Every quadrilateral is a square.

c. Every square is a parallelogram.

d. Every rectangle with four congruent sides is a square.

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Class Name: **ALEKS Integrated Mathematics II** Number of Questions: **24**

Question 1 of 24

$x = 68$

Question 2 of 24

Sum of the interior angle measures $= 1440°$

Question 3 of 24

Number of sides $= 5$

Question 4 of 24

$x = 139$

Question 5 of 24

Measure of each interior angle $= 135°$
Measure of each exterior angle $= 45°$

Question 6 of 24

Number of sides $= 8$

Question 7 of 24

$m\angle E = 104°$
$m\angle EFH = 39°$
$x = 1$

Question 8 of 24

(a) $\triangle JNM \cong \triangle LNK$
(b) $\overline{JN} \cong \overline{LN}$
(c) $\overline{KN} \cong \overline{MN}$
(d)
- ○ Consecutive interior angles of a parallelogram are supplementary.
- ○ Opposite sides of a parallelogram are congruent.
- ○ Opposite angles of a parallelogram are congruent.
- ○ Alternate interior angles formed by the diagonals of a parallelogram are congruent.
- ◉ The diagonals of a parallelogram bisect each other.

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Question 9 of 24

	Statement	Reason
1	$ABCD$ is a parallelogram	Given
2	$\overline{AB} \parallel \overline{CD}$	Definition of a Parallelogram
3	$\overline{AD} \parallel \overline{BC}$	Definition of a Parallelogram
4	$\angle 1 \cong \angle 3$	If lines $\parallel$, then alt. int. $\angle$ s $\cong$
5	$\angle 2 \cong \angle 4$	If lines $\parallel$, then alt. int. $\angle$ s $\cong$
6	$\overline{AC} \cong \overline{AC}$	Reflexive Property
7	$\triangle ABC \cong \triangle CDA$	ASA Congruence Property
8	$\overline{AB} \cong \overline{CD}$ $\overline{AD} \cong \overline{BC}$	CPCTC Property

Question 10 of 24

	Statement	Reason
1	$m\angle A = m\angle C$	Given
2	$m\angle B = m\angle D$	Given
3	$m\angle A + m\angle B + m\angle C + m\angle D = 360°$	Quadrilateral Angle-Sum Property
4	$m\angle A + m\angle B + m\angle A + m\angle B = 360°$	Substitution Property
5	$2(m\angle A + m\angle B) = 360°$	Simplifying
6	$m\angle A + m\angle B = 180°$	Multiplication and Division Properties
7	$\angle A$ and $\angle B$ are supp. angles	Definition of Supplementary Angles
8	$\overline{AD} \parallel \overline{BC}$	If cons. int. $\angle$ s are supp., then lines $\parallel$
9	$m\angle A + m\angle D = 180°$	Substitution Property
10	$\angle A$ and $\angle D$ are supp. angles	Definition of Supplementary Angles
11	$\overline{AB} \parallel \overline{CD}$	If cons. int. $\angle$ s are supp., then lines $\parallel$
12	$ABCD$ is a parallelogram	Definition of a Parallelogram

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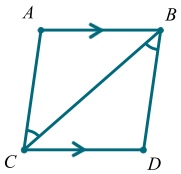
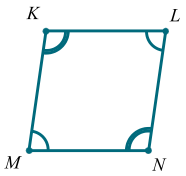
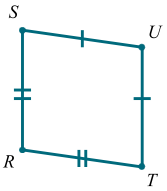
Question 11 of 24



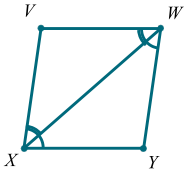
____ Parallelogram



____ Parallelogram



____ Not necessarily a parallelogram



____ Not necessarily a parallelogram

Question 12 of 24

	Statement	Reason
1	$\overline{QR} \cong \overline{ST}$	Given
2	$\overline{QT} \cong \overline{RS}$	Given
3	$\overline{RT} \cong \overline{RT}$	Reflexive Property
4	$\triangle QRT \cong \triangle STR$	SSS Congruence Property
5	$\angle 1 \cong \angle 4$	CPCTC Property
6	$\angle 2 \cong \angle 3$	CPCTC Property
7	$\overline{QR} \parallel \overline{ST}$	If alt. int. $\angle$s $\cong$, then lines $\parallel$
8	$\overline{QT} \parallel \overline{RS}$	If alt. int. $\angle$s $\cong$, then lines $\parallel$
9	$QRST$ is a parallelogram	Definition of a Parallelogram

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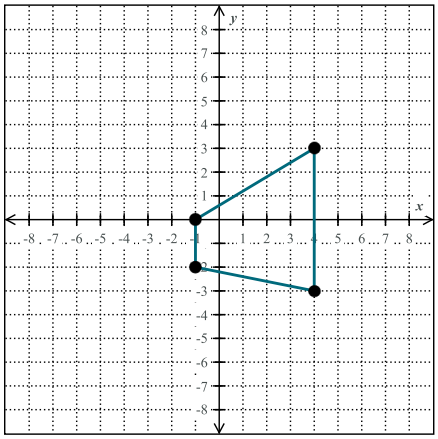
Question 13 of 24

	Statement	Reason
1	$PQRS$ is a parallelogram	Given
2	$\overline{PQ} \parallel \overline{RS}$	Definition of a Parallelogram
3	$\overline{PS} \parallel \overline{QR}$	Definition of a Parallelogram
4	$\angle P$ and $\angle S$ are supp. angles	If lines $\parallel$, then cons. int. $\angle$s are supp.
5	$\angle P$ and $\angle Q$ are supp. angles	If lines $\parallel$, then cons. int. $\angle$s are supp.
6	$\angle Q \cong \angle S$	If $\angle$s are supp. to the same $\angle$, they are $\cong$
7	$\angle R$ and $\angle Q$ are supp. angles	If lines $\parallel$, then cons. int. $\angle$s are supp.
8	$\angle P \cong \angle R$	If $\angle$s are supp. to the same $\angle$, they are $\cong$

Question 14 of 24



Square	Triangle	Trapezoid	None of these
○	○	◉	○

Question 15 of 24

$(2, -3)$

Question 16 of 24

$D(f, 0)$

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Question 17 of 24

(a) Find the slope of $\overline{GK}$ and the slope of $\overline{HJ}$.

Slope of $\overline{GK}$: $-\frac{3}{5}$

Slope of $\overline{HJ}$: $-\frac{3}{5}$

(b) Find the slope of $\overline{JK}$ and the slope of $\overline{GH}$.

Slope of $\overline{JK}$: $\frac{5}{4}$

Slope of $\overline{GH}$: $\frac{5}{4}$

(c) From parts (a) and (b), what can we conclude?

- ○ The quadrilateral is not a parallelogram because it has a pair of opposite sides that are not parallel.
- ○ The quadrilateral is a parallelogram because it has one pair of opposite sides that are parallel, even though the other two sides are not parallel.
- ◉ The quadrilateral is a parallelogram because it has two pairs of opposite sides that are parallel.
- ○ It cannot be determined if the quadrilateral is a parallelogram.

Question 18 of 24

(a) Which figures are parallelograms?
Mark all that apply.

☑ Figure A ☑ Figure B ☐ Figure C ☐ None of the figures

(b) Which figures are squares?
Mark all that apply.

☑ Figure A ☐ Figure B ☐ Figure C ☐ None of the figures

(c) Which figures are rectangles?
Mark all that apply.

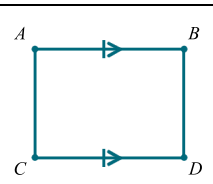
☑ Figure A ☐ Figure B ☐ Figure C ☐ None of the figures

Question 19 of 24

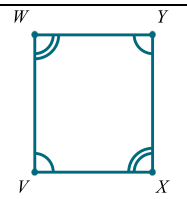
$$LN = 16$$

$$m\angle KNL = 130°$$

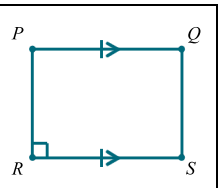
Question 20 of 24



☑ Quadrilateral
☑ Parallelogram
☐ Rectangle



☑ Quadrilateral
☑ Parallelogram
☐ Rectangle



☑ Quadrilateral
☑ Parallelogram
☑ Rectangle

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Question 21 of 24

$m\angle 1 = 33°$
$m\angle 2 = 57°$
$m\angle 3 = 57°$
$m\angle 4 = 33°$

Question 22 of 24

(a) Find the slope of $\overline{XY}$ and the slope of a side adjacent to $\overline{XY}$.

Slope of $\overline{XY}$: $-\frac{9}{4}$

Slope of side adjacent to $\overline{XY}$: $-\frac{5}{4}$

(b) Find the length of $\overline{XY}$ and the length of a side adjacent to $\overline{XY}$. Give exact answers (not decimal approximations).

Length of $\overline{XY}$: $\sqrt{97}$

Length of side adjacent to $\overline{XY}$: $\sqrt{41}$

(c) From parts (a) and (b), what can we conclude about parallelogram $WXYZ$? Check all that apply.

- [ ] $WXYZ$ is a rectangle.
- [ ] $WXYZ$ is a rhombus.
- [ ] $WXYZ$ is a square.
- [x] $WXYZ$ is none of these.

Question 23 of 24

(a) Which properties must be true for parallelograms?

- [ ] Four right angles
- [ ] All sides congruent
- [x] Two pairs of parallel sides
- [ ] Only one pair of parallel sides

(b) Which properties must be true for rectangles?

- [x] Four right angles
- [ ] All sides congruent
- [x] Two pairs of parallel sides
- [ ] Only one pair of parallel sides

(c) Which properties must be true for rhombuses?

- [ ] Four right angles
- [x] All sides congruent
- [x] Two pairs of parallel sides
- [ ] Only one pair of parallel sides

Question 24 of 24

a. Every rhombus is a parallelogram.
True

b. Every quadrilateral is a square.
False

c. Every square is a parallelogram.
True

d. Every rectangle with four congruent sides is a square.
True

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